罗马混凝土的千年之谜

The Secret of Rome's Undying Concrete



古罗马混凝土为何如此耐久？

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RECALL 回想巩固

英文原文

Nearly 1,900 years after its completion, in about AD 125–128, the Pantheon's dome still crowns the vast chamber below. About 43 metres in diameter, it remains the world's largest unreinforced concrete dome. It is an astonishing feat of architecture, and a stubborn puzzle.

Modern concrete is usually reinforced with embedded steel bars, which are prone to corrosion. As the steel rusts, it swells and fractures the concrete around it, and many structures are designed for a lifespan of around 50 years. Roman concrete is not stronger than its modern counterpart; its merit lies in durability.

Yet the ancient material seemed to have a shortcoming. It is dotted with white chunks of lime, known as lime clasts. Experts long ascribed these lumps to sloppy mixing or inferior raw ingredients, a defect that, they assumed, a more conscientious builder would have avoided.

A study published in Science Advances in January 2023 offered a different interpretation. The clasts, it suggested, were a by-product of "hot mixing" with quicklime, a reactive lime that heats up in water. When water seeps into a crack, the clasts dissolve and their calcium recrystallises as calcium carbonate, plugging the gap. In tests, water ceased to trickle through cracked hot-mixed samples within weeks, but kept flowing through samples without quicklime.

The implications extend beyond archaeology. Cement manufacture accounts for about 8 per cent of global carbon dioxide emissions, the think-tank Chatham House reported in June 2018. Concrete that mends its cracks could last longer, making demolition and rebuilding rarer and trimming emissions. The white lumps once scorned as signs of clumsy workmanship have prompted a rethink of construction.

中文译文

万神殿约在公元125至128年竣工。近1900年过去，它的穹顶依然高踞在宽阔的殿堂之上。穹顶直径约43米，至今仍是世界上最大的无钢筋混凝土穹顶。这是建筑史上的惊人壮举，也是一道始终难解的谜题。

现代混凝土通常用埋入其中的钢筋加固，而钢筋容易锈蚀。钢筋生锈后会膨胀，撑裂周围的混凝土；许多建筑的设计寿命约为50年。古罗马混凝土的强度并不比现代混凝土更高，它的优势在于耐久。

不过，古代材料看起来也有一个缺点：内部散布着白色石灰小块，称为石灰碎屑。专家长期把这些团块归咎于搅拌马虎或原料劣质，认为这是一种更认真负责的工匠本应避免的缺陷。

2023年1月，《科学进展》的一项研究提出了另一种解释。研究认为，这些碎屑是使用生石灰进行“热混合”留下的副产物；生石灰活性强，遇水会放热。水渗入裂缝时，碎屑会溶解，其中的钙重新结晶为碳酸钙，堵住缝隙。试验中，开裂的热混合样本在几周内就不再渗水，而不含生石灰的样本仍持续漏水。

这一发现的意义不止于考古。查塔姆研究所2018年6月报告称，水泥生产约占全球二氧化碳排放的8%。能自行修补裂缝的混凝土，可能延长建筑寿命，减少拆除与重建，进而降低排放。那些曾被轻视为手艺拙劣证据的白色团块，如今促使人们重新思考建造方式。

核心词汇

先记这22项。连着下方的原文短语一起回想。

feat n.
壮举；非凡成就
It is an astonishing feat of

prone to adj.phr.
容易发生；易于
which are prone to corrosion.

corrosion n.
锈蚀；腐蚀
which are prone to corrosion.

lifespan n.
使用寿命
for a lifespan of around 50 years.

counterpart n.
对应的事物
stronger than its modern counterpart;

durability n.
耐久性
its merit lies in durability.

shortcoming n.
缺点；不足
material seemed to have a shortcoming.

ascribe to phr.v.
与to连用：归因于
Experts long ascribed these lumps to

inferior adj.
劣质的
mixing or inferior raw ingredients,

conscientious adj.
认真负责的
more conscientious builder would have

interpretation n.
解释
offered a different interpretation.

by-product n.
副产物
were a by-product of "hot mixing" with

quicklime n.
生石灰
of "hot mixing" with quicklime,

reactive adj.
化学活性强的
reactive lime that heats up in water.

seep into phr.v.
缓慢渗入
When water seeps into a crack,

dissolve v.
溶解
the clasts dissolve and their calcium

implication n.
意义；可能影响
The implications extend beyond

extend beyond v.phr.
超出……范围
The implications extend beyond

account for phr.v.
占……比例
Cement manufacture accounts for about

emission n.
排放量
of global carbon dioxide emissions,

demolition n.
拆除
making demolition and rebuilding rarer

prompt v.
促使
have prompted a rethink of

文中词汇与搭配

补充词汇共44项。保留每个表达在本文中的意思。

completion n.
竣工；完成

dome n.
穹顶

crown v.
位于……顶部

vast adj.
宽广的；巨大的

chamber n.
大厅；殿堂

in diameter prep.phr.
直径为

unreinforced concrete n.phr.
无钢筋混凝土

astonishing adj.
令人惊叹的

architecture n.
建筑；建筑艺术

stubborn adj.
难以解决的

puzzle n.
谜题

reinforce with v.phr.
用……加固

embedded adj.
埋入内部的

steel bar n.phr.
钢筋

rust v.
生锈

swell v.
膨胀

fracture v.
使断裂；使开裂

structure n.
建筑物；结构

merit n.
优点；长处

dotted with adj.phr.
散布着

chunk n.
小块；厚块

clast n.
碎屑；碎屑颗粒

文中词汇与搭配 续

补充词汇共44项。保留每个表达在本文中的意思。

lump n.
团块

to prep.
标示所归因的对象

sloppy adj.
马虎的

raw ingredient n.phr.
原材料

defect n.
缺陷

assume v.
认为；未经证实地假定

heat up phr.v.
变热；升温

crack n.
裂缝

calcium n.
钙

plug v.
堵住

gap n.
缝隙

cease to v.phr.
不再；停止

trickle v.
细细流淌

within weeks prep.phr.
在几周之内

archaeology n.
考古学

manufacture n.
生产；制造

think-tank n.
智囊团；研究机构

mend v.
修补

trim v.
削减

scorn v.
轻视

clumsy adj.
拙劣的；笨拙的

rethink n.
重新考虑

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 which are prone to _____.

02 its merit lies in _____.

03 When water _____ a crack,

04 Cement manufacture _____ about 8 per cent of global carbon dioxide emissions,

文章主线

- 耐久 万神殿直径约43米的无钢筋穹顶已存近1900年；罗马混凝土长于耐久，并非强度高于现代混凝土。
- 疑点 现代钢筋锈胀会撑裂混凝土；罗马材料中的白色石灰块，过去却被当成搅拌差或原料劣的缺陷。
- 新解 2023年研究认为，生石灰热混合留下石灰块；水渗入裂缝后，石灰块溶解、钙重新结晶为碳酸钙，堵住裂缝。
- 试验 开裂的热混合样本几周内不再渗水，未加生石灰的对照仍漏水，支持这种自愈解释。
- 前景 水泥生产约占全球二氧化碳排放8%；若自愈材料能延长建筑寿命，就有望减少拆建及排放。

参考答案　1 corrosion 2 durability 3 seeps into 4 accounts for